

Cloud-Only Scripts User Guide (Concise)

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Overview

Seven PowerShell scripts for cloud-only Microsoft Entra ID administration. Each script is CSV-driven, uses Microsoft Graph, includes logging, and handles errors gracefully.

Script	Purpose
Entra_BulkUserImport.ps1	Create users from CSV (licenses, groups, manager)
Entra_Process_Leaver.ps1	Offboard users (disable, convert mailbox, remove access)
Entra_UpdateUser.ps1	Update addresses/phones and other properties
Entra_SetSMTPAddress.ps1	Set primary email (Mail property)
Entra_PhoneAuthentication.ps1	Configure MFA phone number
Entra_Set_Operative.ps1	Set operative/non-operative status (extensionAttribute1)
Entra_ExportAllUsers.ps1	Export all users with all properties to CSV

Prerequisites

- PowerShell 5.1+ (PowerShell 7 recommended)
- Modules: Microsoft.Graph (all scripts), ExchangeOnlineManagement (Leaver only)
- Admin account with appropriate permissions

Install modules:

```
Install-Module Microsoft.Graph -Scope CurrentUser -Force
```

```
Install-Module ExchangeOnlineManagement -Scope CurrentUser -Force
```

Permissions: - Entra_BulkUserImport, Entra_Process_Leaver: User.ReadWrite.All, Directory.ReadWrite.All - Entra_UpdateUser, Entra_SetSMTPAddress, Entra_Set_Operative: User.ReadWrite.All - Entra_PhoneAuthentication: User.ReadWrite.All, UserAuthenticationMethod.ReadWrite.All - Entra_ExportAllUsers: User.Read.All, Directory.Read.All

CSV location: - Default: UserImport.csv in the script directory - Override with
-CsvPath "C:\Path\file.csv"

Entra__BulkUserImport.ps1

Purpose: Bulk create users from CSV with optional licenses, groups, manager, and extension attributes.

Run:

```
.\Entra_BulkUserImport.ps1  
.\Entra_BulkUserImport.ps1 -CsvPath "C:\Path\NewEmployees.csv"
```

CSV (required): First name, Last name, Display name, User principal name, Password, Account status, Company

CSV (common): Job Title, Department, Telephone number, Mobile, E-mail, Manager Email, IsFieldOperative, D365FieldService, Licenses, Groups, StreetAddress, City, Post code, Country

Notes: - Skips existing users - Licenses by SKU name or GUID (Get-MgSubscribedSku)
- Logs to Entra_BulkCreateUsers_YYYYMMDD_HHMMSS.log

Entra__Process_Leaver.ps1

Purpose: Automate offboarding (disable, convert mailbox to shared, clear attributes, remove groups, remove licenses).

Run:

```
.\Entra_Process_Leaver.ps1  
.\Entra_Process_Leaver.ps1 -CsvPath "C:\Leavers\Leavers.csv" -ExchangeUserPrincipalName "ex"
```

CSV: User principal name

Notes: - Exchange Online is optional; leave -ExchangeUserPrincipalName empty to skip - Logs to Entra_Process_Leaver_YYYYMMDD_HHMMSS.log

Entra__UpdateUser.ps1

Purpose: Update user properties (addresses, phones, etc.) from CSV; supports selective updates.

Run:

```
.\Entra_UpdateUser.ps1  
.\Entra_UpdateUser.ps1 -CsvPath "C:\Updates\UserUpdates.csv" -PropertyToUpdate "mobile" # of
```

CSV (minimal): User principal name

CSV (common): Mobile, Telephone number, StreetAddress, City, Post Code, Country

Notes: - Accepts flexible column names (e.g., PostalCode or Post Code) - Logs to `Entra_UpdateUser_YYYYMMDD_HHMMSS.log`

Entra_SetSMTPAddress.ps1

Purpose: Set the default SMTP address (primary email) from CSV.

Run:

```
.\Entra_SetSMTPAddress.ps1  
.\Entra_SetSMTPAddress.ps1 -CsvPath "C:\EmailUpdates\NewAddresses.csv"
```

CSV: User Principal Name, E-mail

Notes: - Updates Mail property (primary SMTP) - Does not set secondary proxy addresses

Entra_PhoneAuthentication.ps1

Purpose: Configure user phone for authentication (MFA).

Run:

```
.\Entra_PhoneAuthentication.ps1  
.\Entra_PhoneAuthentication.ps1 -CsvPath "C:\PhoneSetup\UserPhones.csv"
```

CSV: User Principal Name, Mobile (international format, e.g., +1-555-1234)

Permissions: User.ReadWrite.All, UserAuthenticationMethod.ReadWrite.All

Notes: - If the same number exists, script skips or replaces as needed

Entra_Set_Operative.ps1

Purpose: Update operative status using extensionAttribute1.

Run:

```
.\Entra_Set_Operative.ps1  
.\Entra_Set_Operative.ps1 -CsvPath "C:\Updates\OperativeStatus.csv"
```

CSV: User Principal Name, IsFieldOperative (Operative | Non-Operative)

Entra_ExportAllUsers.ps1

Purpose: Export all users from Entra ID with all properties to CSV (for audit, compliance, analysis, or as baseline for bulk updates).

Run:

```
.\Entra_ExportAllUsers.ps1  
.\Entra_ExportAllUsers.ps1 -OutputPath "C:\Exports\AllUsers.csv"
```

Exports (30+ columns): - Identity: UPN, Display name, First/Last name, Email, Mail nickname, User ID (GUID) - Contact: Mobile, Office phone, Office location, Street address, City, State, Postal code, Country - Organization: Department, Job title, Company name, Manager (resolved) - Account: Enabled status, User type, Employee ID, Created date, Last password change date, On-Premises DN - Custom: Extension Attributes 1–15 - Cloud: Licenses (semicolon-separated), Groups (semicolon-separated)

Notes: - No CSV input required - Creates `EntraUsers_Export.csv` by default - Logs to `Entra_ExportAllUsers_YYYYMMDD_HHMMSS.log` - Updates progress every 50 users - Manager names are resolved from Entra ID - Useful for audits, compliance reports, or as basis for batch updates

Workflows & Best Practices

Onboarding: - Prepare CSV with user details (include Company) - Run `Entra_BulkUserImport.ps1`, then `Entra_PhoneAuthentication.ps1` - Run `Entra_ExportAllUsers.ps1` to verify and audit

Relocation: - Run `Entra_ExportAllUsers.ps1` to export current state - Update address/company columns - Run `Entra_UpdateUser.ps1 -PropertyToUpdate "address"` - Update operative status if needed

Email Change: - Prepare CSV with UPN + new email - Run `Entra_SetSMTPAddress.ps1` - Confirm in Entra admin portal

Offboarding: - Prepare CSV with UPNs - Run `Entra_Process_Leaver.ps1` (Exchange admin optional) - Verify logs and portal

Audit/Reporting: - Run `Entra_ExportAllUsers.ps1` regularly - Review CSV in Excel for compliance - Track changes over time with timestamped exports

Bulk Corrections: - Run `Entra_ExportAllUsers.ps1` to baseline - Identify issues in the export - Update columns and use appropriate script to fix - Re-export to verify corrections

Best Practices: - Test with 1–2 users first - Keep CSVs and logs with timestamps - Use exports for audits and compliance - Export before major changes to have rollback baseline - Review logs after every script run

- Batch large operations and watch API limits
-

Troubleshooting (Quick)

Authentication & Permissions

“Request Authorization failed” Problem: You lack required permissions

Solution: 1. Check your admin role in Entra ID 2. Ensure account has User.ReadWrite.All scope 3. Re-run script and re-consent to permissions 4. Wait 15 minutes for permission changes to propagate

“Connect-MgGraph: Insufficient privileges” Problem: Account lacks required permissions

Solution: 1. Use Global Admin or delegated admin account 2. Verify account has required Microsoft Graph scopes 3. Check Azure AD application registration

“You don’t have permission to view directory” Problem: Account scope too limited

Solution: 1. Request Global Admin or Application Admin role 2. Or request delegated User.ReadWrite.All + Directory.ReadWrite.All 3. Wait for role assignment to take effect

CSV & Data Issues

“CSV file not found: UserImport.csv” Problem: CSV location incorrect

Solution: 1. Place CSV in same directory as script, OR 2. Use -CsvPath parameter: `.\Script.ps1 -CsvPath "C:\Path\File.csv"` 3. Check file name spelling (case-insensitive on Windows)

“User principal name missing” Problem: Required CSV column not found

Solution: 1. Check CSV headers exactly match 2. Ensure no extra spaces before/after 3. Verify column capitalization

“User already exists, skipping” Problem: User with that UPN already in tenant

Solution: 1. Use different UPN in CSV, OR 2. Delete existing user first, OR 3. Use `Entra_UpdateUser.ps1` to update instead

“Company name missing” or “IsFieldOperative missing” Problem: Required field empty in CSV

Solution: 1. Add Company value for all users 2. Add IsFieldOperative value for all users 3. Run script again

User Creation Issues

“License not found: OFFICE365_BUSINESS_PREMIUM” Problem: License SKU incorrect or doesn’t exist

Solution: 1. List available licenses: `powershell Connect-MgGraph Get-MgSubscribedSku | Select SkuPartNumber, SkuId` 2. Update CSV with correct license name or GUID 3. Re-run script

“Group not found: Engineering Team” Problem: Group doesn’t exist in tenant

Solution: 1. Create group first in Entra ID 2. Verify group name spelling in CSV 3. Re-run script

“Manager not found: manager@contoso.com” Problem: Manager user doesn’t exist

Solution: 1. Verify manager email/UPN is correct 2. Ensure manager user exists in tenant 3. Fix CSV and re-run

Update Issues

“Property not found” (Entra_UpdateUser) Problem: CSV column name incorrect

Solution: 1. Use exact column names from documentation 2. Script accepts both “Post Code” and “PostalCode” 3. Check for extra spaces in headers

User properties not updated Problem: PropertyToUpdate parameter wrong

Solution: 1. Use valid values: “all”, “mobile”, “address” 2. Example: `-PropertyToUpdate "mobile"` 3. Default is “all” if not specified

Phone Authentication Issues

“Phone already exists” (Entra_PhoneAuthentication) Problem: User already has this phone method

Solution: 1. Script skips if phone already set 2. If updating, script removes old and adds new 3. Check log for details

“Invalid phone format” Problem: Phone number not in correct format

Solution: 1. Use international format: +1-555-1234 2. Include country code 3. Format: +[country code]-[area code]-[number]

Offboarding Issues

“Exchange connection failed” Problem: Exchange Online credentials invalid

Solution: 1. Verify Exchange admin email/UPN 2. Ensure account has Exchange admin role 3. Leave empty to skip mailbox conversion: "" 4. Run PowerShell Console (not ISE)

“Mailbox conversion failed” Problem: Mailbox couldn’t be converted to shared

Solution: 1. Check user has a mailbox 2. Ensure Exchange admin permissions 3. Try converting manually in Exchange admin portal 4. Check log for specific error message

“Group removal failed” Problem: User not member of group

Solution: 1. Script logs and continues (non-fatal) 2. Check log for specific groups that failed 3. Manually remove if needed

Module Issues

“Module Microsoft.Graph not found” Problem: Graph module not installed

Solution: 1. Script installs automatically on first run 2. If fails, install manually: `powershell Install-Module Microsoft.Graph -Scope CurrentUser -Force`

“ExchangeOnlineManagement not found” Problem: Exchange module not installed (Leaver script only)

Solution: 1. Install manually: `powershell Install-Module ExchangeOnlineManagement -Scope CurrentUser -Force` 2. Or skip mailbox conversion in `Leaver_Script`

PowerShell Issues

“Execution Policy” Problem: Scripts won’t run due to execution policy

Solution:

`Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser`

“Variable reference is not valid” Problem: String interpolation error in script

Solution: 1. Should not occur in updated scripts 2. If you encounter this, contact support 3. Error appears as: “‘.’ was not followed by a valid variable name”

Script pauses at end Problem: “Pause” command at end of script

Solution: 1. This is normal - press Enter to close 2. Or remove “Pause” line from script if batch running

Support

Contact: Peter James (peter.james@nexusos.co.uk)

Include: script name, error message, CSV sample, PowerShell version, steps to reproduce, and log file.

Resources: - [Microsoft Graph PowerShell SDK](#) - [Entra ID Docs](#) - [Exchange Online PowerShell](#) (for mailbox tasks)
